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Understanding effects of process parameters and forced feeding on die filling

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ABSTRACT

Die filling is a critical step during pharmaceutical tablet production and is still not well understood due to the rather complex interplay between particle attributes, die orifice diameter and fill energetics. While shoe-die filling models have been used to simulate die filling conditions, they typically lack the sophistication of the actual production-scale, feeder-based die filling conditions. The relationship between tableting process parameters and filling into die orifices of different diameters by powders of different flowabilities requires critical examination and understanding. In this study, a special die filling contraption was designed and custom-made to simulate the effects of gravity, suction and feeder paddle assistance as present in modern rotary tablet presses. Die fill performance was studied using powders with different flow properties. Suction impact was greatest on die fill, in particular, for small orifice diameters and less permeable powders. Effect of paddle velocity on die fill was greater for compressible powders and larger orifice diameters. In comparison to suction and paddle velocity, forced feeding did not significantly affect die fill performance. Relationship between process parameters and die fill performance was found to be highly dependent on the material and orifice diameter.

1. Introduction

Production of pharmaceutical tablets typically involves the three consecutive process steps of die filling, compression and tablet ejection. The die fill determines tablet weight which in turn is related to drug content and other critical quality attributes such as tablet mechanical strength, friability and disintegration. Reproducibility of die fill is hence critical as it not only affects product quality but also the overall efficiency of the tableting process (Mills and Sinka, 2013).

Research in die fill was initially driven by other industries involved in powder compaction, such as powder metallurgy and ceramics (Bocchini, 1987). The die filling process can be passive (powder delivered into a stationary die from a moving shoe) or active (powder delivered into a moving die from a stationary shoe) on the compaction system (Peeters et al., 2015; Wu, 2008). By using shoe-die models to examine the passive die fill process, interactions between powder and air during die filling could be elicited. Displacement of air from the die cavity by powder filling created a pressure gradient which opposed further powder entry into the cavity and this could be visualised by high speed video (Mills and Sinka, 2013; Wu and Cocks, 2004; Xie and Puri, 2006). Higher die fill densities achieved by the assistance of vacuum pointed to the adverse effect of the presence of air in the die hindering its filling (Wu et al., 2003). It has been found that gravity fill of powder into a die cavity is primarily governed by three types of flow regimes: nose flow, bulk flow and intermittent flow (Mills and Sinka, 2013; Sinka et al., 2004; Wu et al., 2003; Wu and Cocks, 2004). As the feed shoe moves across the die cavity, the powder adopts a nose-shaped profile due to inertia and frictional interaction between the powder and the surface on which the shoe slides. The initial cascading of particles down the slope into the die cavity is termed as nose flow. Bulk flow occurs when powder detaches smoothly from the bottom of the powder bed into the die after the feed shoe has completely covered the die cavity. In cases where powder agglomerates detach by random into the die, the flow is termed as intermittent flow.

The die fill for shoe-die model studies is usually quantified by fill ratio and feed shoe critical velocity. Fill ratio refers to the ratio between the powder mass collected in the die after one pass of the feed shoe and the powder mass in a completely filled die (Mills and Sinka, 2013). It can also be the ratio between the volume of powder collected in the die to the die volume (Wu, 2008). A higher fill ratio would mean higher fill density inside the die. The concept of critical velocity (maximum feed shoe velocity that can fill the die cavity completely with a single pass) was also used by researchers as an indicator of powder flowability and die fill performance (Mills and Sinka, 2013; Sinka et al., 2004). High fill ratio and critical velocity indicate better die fill performance.

The shoe-die model was found to underestimate powder flow performance by half when up-scaling the data to a rotary tablet press (Schneider et al., 2007). The unaccounted suction fill effect present in rotary presses was identified subsequently through a modified shoe-die model with a movable lower punch that descended just as the feed shoe

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translated across the die cavity (Jackson et al., 2007). Suction fill created a negative pressure gradient and eliminated the effects of air entrainment encountered during gravity fill. Under these conditions, higher packing density and lower risk of segregation during die fill were achieved (Mills and Sinka, 2013).

Regardless of the type of fill regime, die filling is primarily affected by three factors: powder properties, die characteristics and feeding mechanism (Peeters et al., 2015; Xie and Puri, 2006). Under conditions of gravity fill, large and dense particles had higher die filling rates and critical velocities than fine particles as air expelled more easily from the die cavity through coarser particles (Mills and Sinka, 2013; Wu et al., 2003). Powder systems with wider particle size distributions also showed higher filling rates and critical velocities (Wu, 2008). The effect of die geometry on the die filling process has also been studied. Net die filling rate was slower in a stepped die compared to a simple die (Wu et al., 2003). Critical velocity was also found to increase with wider die diameter or shorter die height (Sinka et al., 2004). For dies with very small widths, gravity fill of powder occurred mainly via bulk flow due to insufficient time for nose flow to occur (Schneider et al., 2007).

Die filling on rotary presses is usually an active process with powder flowing from a stationary feed frame into moving dies on the rotating turret (Wu, 2008). The feed frame can contain up to three chambers in which coplanar paddle feeders rotate continuously to circulate the tablet feed and push it into the dies through a process known as forced feeding (Peeters et al., 2015; Wu, 2008). Variations of the feed frame design in terms of number, size and shapes of paddles are available depending on the tablet press model (Gopireddy et al., 2016). High paddle speed was associated with increased die fill weights and reduced die fill variation (Mendez et al., 2010). This was attributed to the higher frequency of paddle passes to push powder into the dies as paddle speed increased. On the other hand, Peeters et al. (2015, 2016) found that the effect of paddle speed on tablet weight was dependent on the powder flow properties. Paddle speed was only positively correlated with tablet weight for poor flowing powders due to the greater forced feeding effect and did not impact tablet weight variation. In contrast, simulations of die filling process on rotary tablet press found that paddle design had only impacted tablet weight variation but not tablet weight (Ketterhagen, 2015).

While various techniques have been used to assess powder flowability in relation to die fill, it is unclear if they are able to reliably predict die filling (Freeman and Fu, 2008; Monedero Perales et al., 1996; Yaginuma et al., 2007). Shoe-die models ranked powder flowability differently from standard flow measurement techniques such as angle of repose and bulk/tapped density measurements (Mills and Sinka, 2013). Furthermore, the interplay of gravity fill (flow of powder into die by gravity), suction fill (partial vacuum created by descent of lower punch at fill cam draws powder into die) and forced feeding (rotation of paddles in feed frame assist powder flow into die) in a rotary tablet press makes the study of the actual die filling process very challenging. The shoe-die model is possibly the closest at mimicking conditions on a rotary press albeit with limitations (Schneider et al., 2007). Ideally, a “die-filling” system should be able to reproduce the effects of gravity fill, suction and forced feeding, with the flexibility of different die configurations, paddle designs and feeder velocities. It is also unclear how powders with different flowability would respond to these effects. The purpose of this study was twofold, firstly, to design an apparatus capable of performing gravity fill, suction and forced feeding of powders for different die diameters and secondly, to examine the effects of paddle design, paddle velocity and suction on die fill of three representative powders using the designed system.

2. Materials and methods

2.1. Materials

Nonpareils (NP0; Suglets®, Colorcon®, USA) of 250–355 μm size fraction were used as a model coarse and free flowing powder. Directly compressible lactose (T100; Tablettose 100 M, Meggle, Germany) was used as a fine and fair flowing powder. Magnesium stearate (M125, Productos Metalast, Spain) was employed as the lubricant.

2.2. Preparation of lubricated lactose powder

Lubricated lactose powder (T100M) was prepared by blending T100 with 1%, w/w magnesium stearate in a double cone blender (AR401, Erweka, Germany) rotated at 15 rpm for 5 min. The total blended mass was 500 g. T100M was used as a powder with flow properties intermediate between that of NP0 and T100.

2.3. Characterisation of particle size

Particle size and size distribution of NP0 and T100 were determined by a laser diffractometer (LS230, Beckman Coulter, USA) with the dry powder module and a target obscuration level of about 10%. The vacuum source for the dry powder module was preset to draw air in at 2.83 L/min. A cumulative undersize curve was generated from which particle size parameters corresponding to the 10th, 50th and 90th percentiles (D_{10}, D_{50} and D_{90}) were determined. Span, an indicator of particle size distribution, was calculated using Eq. (1).

$$\text{Span} = \frac{D_{90} - D_{10}}{D_{50}} \tag{1}$$

2.4. Flow characterisation

2.4.1. Determination of angle of repose

Approximately 100 g of sample was passed through a funnel onto a circular base plate (Copley Scientific, United Kingdom) of diameter, 2r, so that it formed a conical heap. Height of the funnel was adjusted so that the orifice was about 2–4 cm away from the expected tip of the formed powder heap. Angle of repose (AoR) was calculated from the height, h, of the cone using Eq. (2). Measurements were triplicated and averaged results reported. Low AoR values are usually indicative of good powder flow.

$$\text{AoR} = \tan^{-1}\frac{h}{r} \tag{2}$$

2.4.2. Compressibility test

Powder samples were sifted through a 1 mm aperture size sieve into an exactly 100 mL graduated cylinder. After scraping away the excess powder carefully from the top of the cylinder and dusting off adhering powder, the cylinder was transferred onto a tapping machine (JEL STAV II, J. Engelsmann AG, Germany) and tapped at 250 drops/min until a constant tapped volume. Bulk (ρ_{bulk}) and tapped (ρ_{tapped}) densities were calculated using Eqs. (3) and (4), respectively.

$$\rho_{bulk} = \frac{\text{Sample mass}}{100\ \text{mL}} \tag{3}$$

$$\rho_{tapped} = \frac{\text{Sample mass}}{\text{Tapped volume}} \tag{4}$$

Hausner ratio (HR) was calculated using Eq. (5). The measurements were triplicated with fresh powders and the results averaged. Low HR values are usually associated with good powder flow properties.

$$\text{HR} = \frac{\rho_{tapped}}{\rho_{bulk}} \tag{5}$$

2.4.3. Shear cell test

A powder rheometer (FT 4, Freeman Technology, United Kingdom) fitted with the rotational shear cell accessory was used for shear cell

testing of NP0, T100 and T100 M. Pre-shearing of the samples was performed at a normal load of 9 kPa and repeated before determination of every yield point. A total of five yield points were determined at decreasing normal stresses (7.0, 6.0, 5.0, 4.0 and 3.0 kPa) to construct the yield locus from which powder cohesion (COH) and unconfined yield strength (UYS) were determined. Low COH and UYS values are usually indicative of non-cohesive powders.

2.4.4. Avalanche flowability

Approximately 100 cm^3 of sample was rotated in 10 cm diameter drum at 0.6 rpm and a digital camera captured images of the particulate flow behaviour in the drum as it rotated (Revolution® Powder Analyzer, Mercury Scientific Inc., USA). The drum was initially rotated for 60 s to establish a more consistent flowing powder mass in the drum before the camera captured images of the avalanche behaviour for a total of 150 avalanches. During the run, avalanche angle (AvA; angle of powder inclination just before avalanche event), avalanche time (AvT; time between consecutive avalanche events), and avalanche energy (AvE; amount of energy released by the powder during the avalanche event) were monitored by the software and average values were calculated. Low values of AvA, AvT and AvE are usually indicative of good powder flow.

2.5. Determination of powder permeability

Air permeability describes how easily a material can transmit air through its bulk and can be measured by the pressure drop across the powder bed, which represents the resistance of air to flow through the powder. Materials which are more permeable will have a lower pressure drop. The powder rheometer (FT 4, Freeman Technology, United Kingdom) fitted with an aeration control unit was used for the permeability test. After a conditioning phase, a vented piston applied normal stresses in increasing order (0.2 kPa and 0.5–5 kPa in increments of 0.5 kPa) on the powder sample in the vessel with air passing through the vessel from the bottom at a constant flow rate of 2 mm/s. Each normal stress was applied for a preset duration and the pressure drop across the powder bed determined, which was then used to calculate powder permeability. The average powder permeability across the 11 normal stresses was reported. Under application of normal stress, fine and cohesive powders are typically less permeable than granular materials due to tighter packing of particles and less void spaces present.

2.6. Die filling apparatus

The apparatus designed to study die fill was fabricated as described. Various three-dimensional prototype models were created using computer-aided design software (Solidworks® 2016, Dassault Systems, USA) and the ideal design was chosen for fabrication. The apparatus comprised a mounted 20 mm thick aluminium platform with four equidistant, circular holes representing die orifices of diameters 2, 4, 7 and 10 mm, drilled through the entire thickness of the platform (Fig. 1A). The die orifices (each 20 mm deep) were aligned to each other at their centers, each with one powder collection cup (cup volume: approximately 24,900 mm^3) mounted directly below the platform. Actuation of a feeder unit to traverse across the die orifices was accomplished by means of a rotor connected to the feeder via a metal cord. Speed of the rotor could be adjusted to achieve different linear velocities of the feeder. The feeder unit comprised a main metallic frame and four swappable paddle inserts (Fig. 1B), each aligned to one individual orifice when the feeder was in position on the platform. Paddle inserts with different angles of inclination were available to simulate paddle feeders designed with different extents of forced feeding. Additionally, simulation of suction effect was possible since differential pressures can be applied across the die orifices by means of vacuum suction effected via the collection cup. The greater the differential pressure, the greater is the suction effect.

By closely reproducing some features of the industrial process on rotary tablet presses, the filling apparatus designed could potentially provide conditions to study the more relevant powder flow behaviour from a die fill perspective beyond the capabilities of other flow measurement methods.

2.7. Evaluation of die fill performance

Die fill performance of NP0, T100 and T100M for each of the four die orifices under different process conditions was evaluated: feeder velocity (4, 8 and 12 m/min), paddle angle (30, 45, 60 and 90°) and differential pressure (0.0, 0.1 and 0.2 bar with respect to atmospheric air pressure (Table 1). A higher differential pressure indicates greater negative pressure and suction effect. The chosen feeder velocities were close to the range of feed wheel linear velocities on rotary presses reported in other studies (Goh et al., 2017; Peeters et al., 2015; Peeters et al., 2016). Fig. 1C summarises the operation of the die filling apparatus. Powder sample was introduced into the feeding trench in small amounts until full and excess powder was carefully scraped off, levelling the feed such that a consistent powder head height was obtained since the overhead pressure is known to affect powder flow and die fill (Peeters et al., 2015). Starting position of the feed frame on the platform was fixed so that a consistent time for acceleration from start to fill was maintained. After the sample-loaded feeder unit had traversed across the platform, mass of powder collected in the cup below the orifice was determined.

Paddle inserts with smaller angles had larger trench volumes which would affect the amount of powder available for die fill (Fig. 2). As such the mass of powder collected in the cup for the smaller paddle angles were scaled using the ratio between the respective trench volume and that of the 90° paddle. The distance between the bottom part of the paddle insert and trench front was also longer with decreasing paddle angle which entailed a longer time of contact between the powder and die orifice. Mass flow rate of the powder was calculated using Eq. (6) by considering the scaling factor and the average time taken for the bottom of the paddle to completely cross the die orifice.

$$\text{Mass flow rate} = \frac{\text{Collected powder mass}}{\text{Time}} \div \text{Scaling Factor} \quad (6)$$

Normalising mass flow rate against orifice cross-sectional area enabled comparisons to be made between the different orifice diameters under similar feeding conditions. Mass flow rate per unit orifice area (MFA) is an indicator of die fill performance and was calculated using Eq. (7). The unit of measure for MFA was $g{\cdot}cm^{-2}s^{-1}$. Higher MFA values indicated better die fill performance. For each experimental configuration, six replicate measurements were taken and the results averaged.

$$\text{MFA}_{\#} = \frac{\text{Mass flow rate}}{\text{Orifice cross} - \text{sectional area}} \quad (7)$$

where # represents the die orifice diameter in mm.

Visual observations of the powder orifice flow under static conditions were also made. The orifices were first covered by a thin plastic film with the trenches positioned directly above them. Powder samples were fed into the trenches as per normal. The plastic film was then removed quickly by pulling it away using a rotor operated at linear speed of 12 m/min. Flow behaviours of the powder into the orifices were observed and described qualitatively.

2.8. Statistical analyses

A general full factorial experimental design (Table 1) was created using statistical software (Minitab® 17.1.0, Minitab Inc., USA). All main effects and estimable two-way interactions were analysed with the same software. Partial least squares regression (PLSR) modelling was

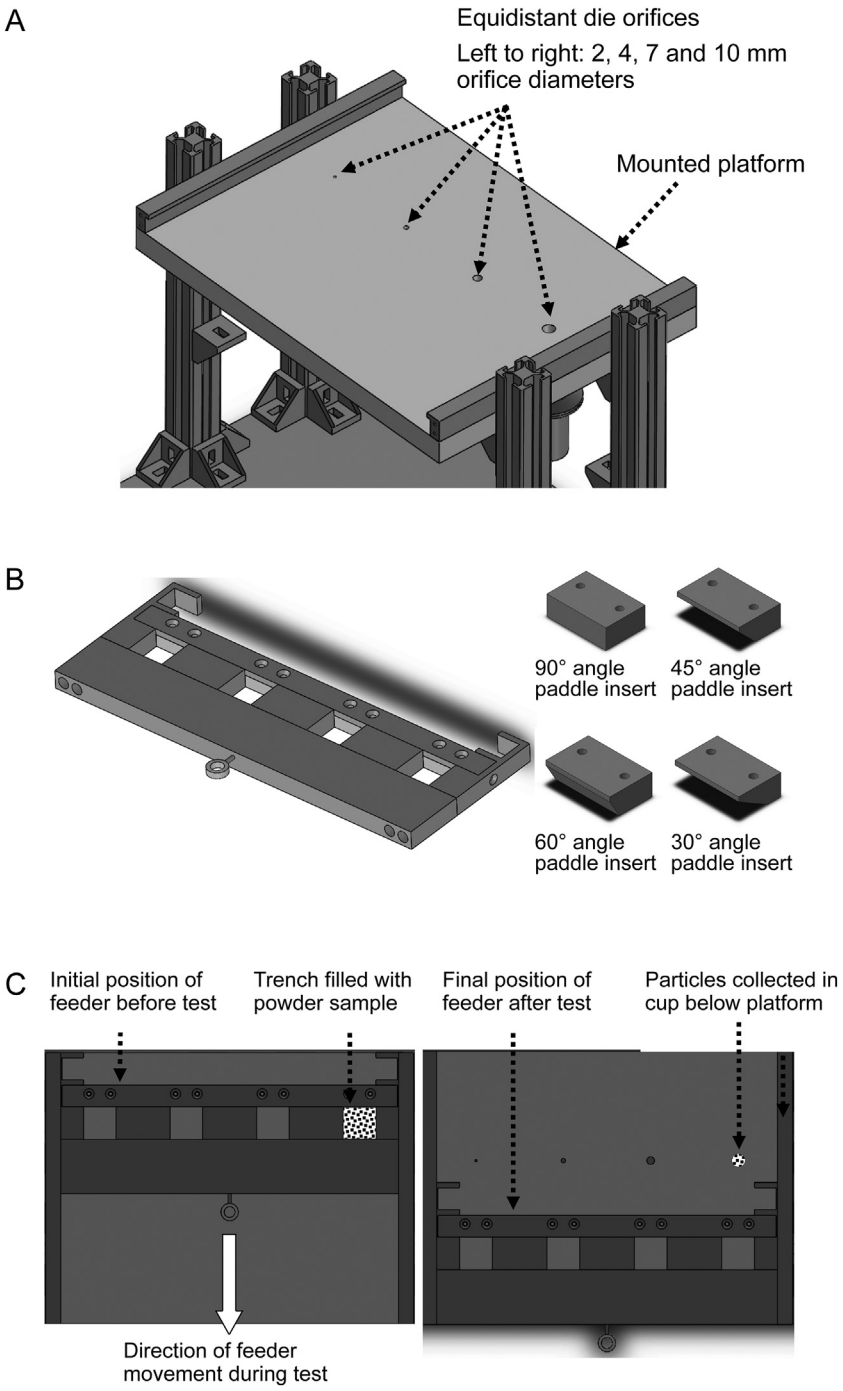


Fig. 1. (A) Mounted platform of die fill device showing the four die orifices. One collection cup was attached below each orifice. (B) Feeder with swappable paddle inserts that have different angles of inclination. (C) Trench in feeder was filled to equal powder head height with sample. Actuation of feeder across platform caused some sample to enter the collection cup via the orifice.

Table 1
Factors and levels used in the general full factorial design.

Factor	Levels
Material	NP0, T100 and T100M
Feeder velocity (m/min)	4, 8 and 12
Paddle angle (°)	30, 45, 60 and 90
Differential pressure (bar)	0, 0.1 and 0.2

also performed in order to determine the relative importance of each process variable on die fill performance (The Unscrambler® X v.10.3, CAMO Software, Norway). Data scaling and cross validation were done prior to regression analyses. The regression models were evaluated based on their goodness-of-fit (R^2) and goodness-of-prediction (Q^2) values.

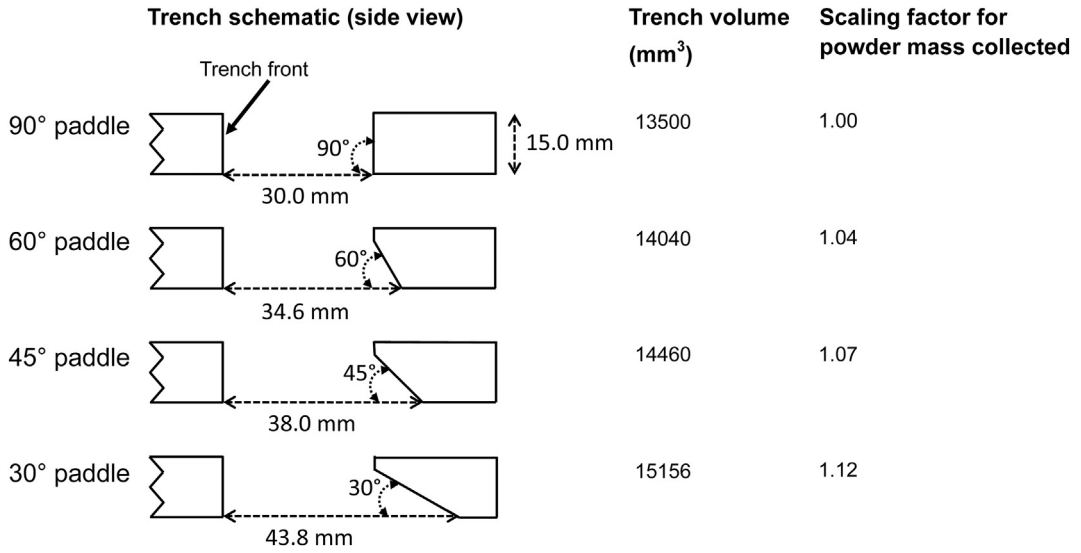


Fig. 2. Schematic of trenches of die filling contraption with corresponding volumes and scaling factors for different paddle angles.

3. Results

3.1. Physical properties of powders

Mean particle size, particle size distribution, bulk densities, flow properties and permeability of the powders are summarised in Table 2. In general, the three powders could be ranked in ascending order of flowability: T100 < T100M < NP0 which corresponded to decrease in AoR, HR, UYS and COH across the powders. NP0 had larger particle and bulk density compared to T100 and T100M which accounted for its better flow properties. Addition of magnesium stearate improved T100 flow properties likely as a result of reduced interparticulate friction (Morin and Briens, 2013).

Interestingly, T100 appeared to perform better than T100M in terms of avalanche flow. This was possibly related to higher avalanche frequency observed in T100 associated with formation of loose agglomerates during the avalanche measurement. This in turn lowered AvA, AvE and AvT and could give a false interpretation without visual observations of the avalanche flow (Soh et al., 2006; Tay et al., 2017; Thalberg et al., 2004).

With regard to permeability, NP0 was the most permeable, followed by T100 and T100M (Table 2). As the three powders had displayed different physical and flow properties, they were suitable for further investigations into die filling.

3.2. Reproducibility of results obtained from die filling apparatus

Mean powder masses collected in the cups and corresponding relative standard deviations (RSD) are summarised in Table 3. With the exception of T100 at 2 mm orifice diameter configuration (RSD: 12.12%), most of the RSDs were below 10%, suggesting that variations were relatively low and results were reproducible. Thus, results obtained from the die filling apparatus were sufficiently reliable for further analyses. NP0 had the lowest RSD, followed by T100M and T100, which could be attributed to their flow properties. Additionally, as orifice diameter decreased from 10 to 2 mm, RSD values generally increased across all the three powders, indicating the greater challenge in achieving fill uniformity for the smaller orifices.

3.3. Powder orifice flow under static conditions

Under static conditions, the following flow behaviours were observed across all the orifice diameters; mass flow for NP0, ratholing for T100M and arching for T100 (Fig. 3). T100 had the poorest flow properties and exhibited arching behaviour even for the largest orifice diameter of 10 mm. On the other hand, NP0 produced mass flow behaviour even for the narrowest orifice diameter of 2 mm. While addition of magnesium stearate had improved powder flow, ratholing was still observed for T100M across all orifice diameters. The orifice flow behaviours had thus corroborated with the powders' flow properties.

3.4. Effects of process parameters on die fill of NP0

Fig. 4 shows the main effects plots of feeder velocity, paddle angle and differential pressure on MFA of NP0, T100 and T100M. Each main effects plot displays the mean response values for each group within a categorical variable. For each factor, when the line connecting the points is horizontal (parallel to the x-axis), it indicates that there is no main effect present since the mean response is same across all factor levels. On the other hand, a main effect is present when the connect line is not horizontal; the mean response is not the same across all factor levels. The steeper the gradient of the connect line, the greater the magnitude of the main effect.

Across the different orifice diameters, all of the process parameters had significant main effects on MFA of NP0; differential pressure had the greatest impact while feeder velocity and paddle angle were comparable. Higher levels of feeder velocity, paddle angle and differential pressure increased MFA. As orifice diameter increased, the effect of

Table 2
Physical properties of NP0, T100 and T100M.

Material	D_{50} (μm)	Span	ρ_{bulk} (g/cm^3)	AoR (°)	HR	UYS (kPa)	COH (kPa)	AvT (s)	AvE (kJ/kg)	AvA (°)	Permeability (10^9 cm^2)
NP0	295.9 (0.2)	0.34	0.81 (0)	24.2 (0.1)	1.09 (0)	0.58 (0.09)	0.16 (0.03)	1.63 (0.06)	9.31 (0.12)	33.2 (0.1)	768.1 (25.3)
T100	166.8 (5.1)	1.90	0.58 (0.02)	41.0 (0.1)	1.38 (0.04)	1.21 (0.10)	0.31 (0.02)	2.63 (0.06)	24.21 (0.59)	57.4 (1.2)	111.2 (6.3)
T100 M	144.5 (6.4)	2.07	0.68 (0.02)	36.9 (0.3)	1.18 (0.03)	0.71 (0.30)	0.18 (0.08)	3.27 (0.15)	34.02 (1.42)	58.4 (0.3)	68.6 (1.6)

Parenthesised values indicate the respective standard deviations.

Table 3
Average collected powder masses and RSDs across different die orifice diameters.

	Die orifice diameter (mm)							
	2		4		7		10	
Material	Average powder mass (g)	RSD (%)	Average powder mass (g)	RSD (%)	Average powder mass (g)	RSD (%)	Average powder mass (g)	RSD (%)
NP0	0.39	5.37	1.46	2.50	2.91	2.87	4.25	2.49
T100	0.55	12.12	1.32	4.91	1.76	6.60	2.06	5.26
T100M	0.67	7.03	1.44	4.52	1.93	5.31	2.39	4.94

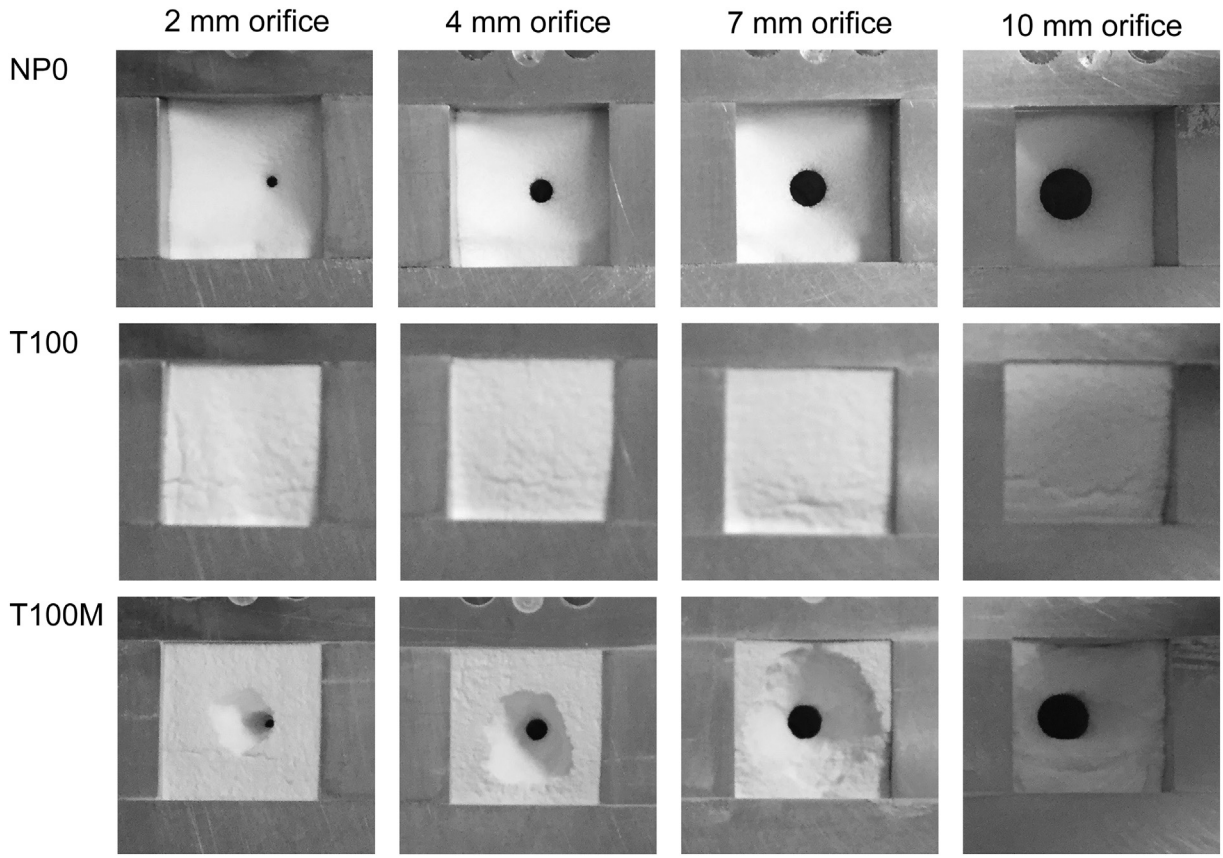


Fig. 3. Orifice flow behaviours for under static conditions.

differential pressure on MFA decreased sharply as seen from the less steep increase in MFA with increasing differential pressure. The increase in MFA between 0 and 0.1 bar was also much larger than that from 0.1–0.2 bar especially for MFA_2 and MFA_4. On the other hand, feeder velocity and paddle angle has almost linear effects on MFA.

Higher overall mean MFA values, as indicated by the dotted horizontal lines in the graphs, were obtained for smaller orifice diameters (MFA_2 – 40.4 $g{\cdot}cm^{-2}s^{-1}$, MFA_4 – 37.1 $g{\cdot}cm^{-2}s^{-1}$, MFA_7 – 24.2 $g.cm^{-2}s^{-1}$ and MFA_{10} – 17.8 $g{\cdot}cm^{-2}s^{-1}$). This could be attributed to the higher sensitivity of smaller orifices to differential pressure as an increase in differential pressure from 0 to 0.1 bar increased MFA_2 by almost 50 $g{\cdot}cm^{-2}s^{-1}$ (Fig. 4A) but only increased MFA_{10} by about 22 $g{\cdot}cm^{-2}s^{-1}$ (Fig. 4D). Higher sensitivity of smaller orifices to differential pressure had thus raised their overall MFA values.

It was interesting to observe that smaller paddle angles had a detrimental effect on MFA. While the mass of powder collected in the cup increased slightly with decreasing paddle angles (Fig. 5A), this slight improvement was not significant ($p > 0.05$). Normalising the powder mass against larger trench volume and longer time of travel across the die orifice for the smaller paddle angles reduced their MFA values.

3.5. Effects of process parameters on die fill of T100

Similar to that of NP0, MFA values of T100 increased with higher levels of feeder velocity, paddle angle and differential pressure (Fig. 4). It was also clear that differential pressure had greater impact on MFA of T100 than feeder velocity and paddle angle, which was in line with observations made for NP0. While increasing differential pressure from 0 to 0.1 bar increased MFA to a greater extent than from 0.1–0.2 bar, both feeder velocity and paddle angle showed relatively linear effects on T100 MFA. Similar to that of NP0, changing the paddle angle did not appear to have significant effect on the powder mass collected for T100 (Fig. 5B). Scaling MFA against the larger trench volume and longer time of contact between powder and die orifice for the smaller paddle angles could explain the observation that MFA increased with paddle angle.

Overall mean values were as follows: MFA_2 – 57.1 $g{\cdot}cm^{-2}s^{-1}$, MFA_4 – 36.7 $g{\cdot}cm^{-2}s^{-1}$, MFA_7 – 16.5 $g{\cdot}cm^{-2}s^{-1}$ and MFA_{10}–9.6 $g{\cdot}cm^{-2}s^{-1}$. Higher mean MFA values were seen for smaller orifices. Interestingly, mean MFA_2 value of T100 was even higher than that of NP0 (Fig. 4A), suggesting that T100 had outperformed NP0 in terms of die fill performance for the narrow orifice despite it possessing relatively poorer flow properties and lower bulk density.

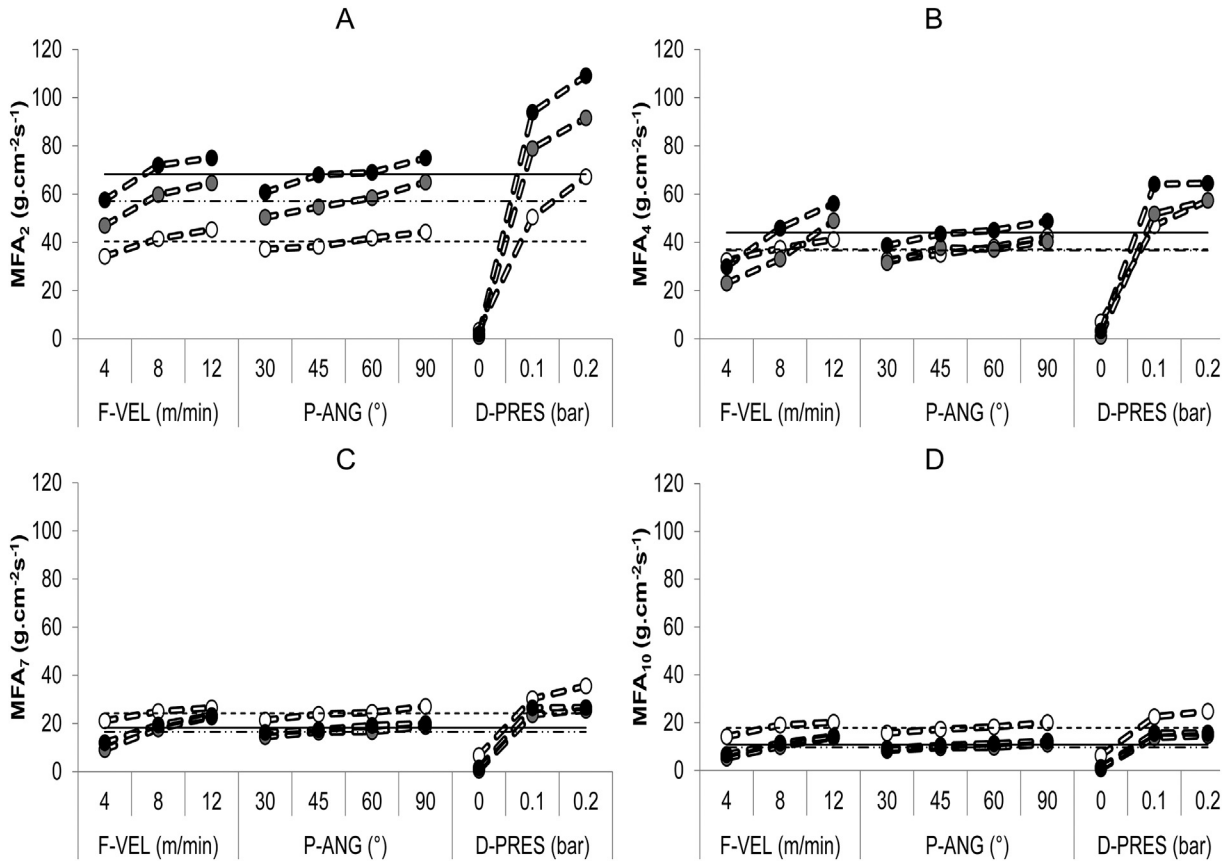


Fig. 4. Main effects plots of feeder velocity (F-VEL), paddle angle (P-ANG) and differential pressure (D-PRES) on (A) MFA_2, (B) MFA_4, (C) MFA_7 and (D) MFA_{10}. Materials: (○) NP0, (●) T100 and (●) T100M. Dotted, dash dot and solid horizontal lines indicate overall means for NP0, T100 and T100M, respectively.

3.6. Effects of process parameters on die fill of T100 M

Overall mean MFA for T100 M values were as follows: MFA_2 – 68.3 $g{\cdot}cm^{-2}s^{-1}$, MFA_4 – 44 $g{\cdot}cm^{-2}s^{-1}$, MFA_7 – 18.2 $g{\cdot}cm^{-2}s^{-1}$ and MFA_{10} – 10.8 $g{\cdot}cm^{-2}s^{-1}$. Observed trends were generally very similar to those T100 with the exception that MFA_2 and MFA_4 of T100 M were higher than those of T100 ($p < 0.001$, Fig. 4A–B). Paddle angle did not appear to have significant impact on the average powder masses collected for T100 M (Fig. 5C). Despite possessing poorer flow than NP0, MFA_2 and MFA_4 values of T100 M were higher than those of NP0.

3.7. Interactions between process variables on die fill

Two-way interactions between feeder velocity, paddle angle and differential pressure were evaluated. Only interaction plots for 10 mm orifice for each material are presented in Fig. 6 since similar observations could be made for the other orifice diameters. In general, if the plotted lines are not parallel to each other, interactions exist; the more nonparallel the lines are, the greater the strength of the interaction.

3.7.1. Feeder velocity and differential pressure

For NP0, increase in feeder velocity at 0 bar differential pressure (gravity fill) decreased MFA_{10} slightly (Fig. 6A(i)). However, at 0.1 and 0.2 bar differential pressures (suction fill), MFA_{10} increased with feeder velocity. The diverging lines between gravity and suction fills suggested strong and significant interaction between feeder velocity and differential pressure ($p < 0.001$). Similarly, while MFA_{10} of T100 and T100 M increased with feeder velocity at 0.1 and 0.2 bar differential pressures, feeder velocity did not appear to have as much effect at 0 bar (Fig. 6B(i) and C(i)) which also point to significant and strong interactions between feeder velocity and differential pressure ($p < 0.001$ for both T100 and T100M).

3.7.2. Feeder velocity and paddle angle

Across all 3 materials tested, an increase in feeder velocity generally increased MFA_{10} regardless of paddle angle used (Fig. 6A(ii)–C(ii)). It could also be seen that increasing paddle angle resulted in progressive upwards shift of the plots from 30° to 90° paddle angles, suggesting that the effect of changing paddle angle was similar regardless of feeder velocity and hence, little interaction between feeder velocity and paddle angle was seen ($p > 0.05$ for all three materials).

3.7.3. Paddle angle and differential pressure

For all three materials, increase in paddle angle increased MFA_{10} at 0.1 and 0.2 bar differential pressure but did not have apparent effect at 0 bar (Fig. 6A(iii) – C(iii)). The diverging lines between gravity and suction fills suggest significant interactions between differential pressure and paddle angle ($p < 0.001$ for all materials).

3.8. Relative importance of process variables on die fill

PLSR modelling was used to evaluate the relative importance of process variables on MFA for each orifice diameter. Paddle angle, feeder velocity and differential pressure were used as predictor variables while MFA was the response variable. Relatively good fits of the datasets and predictability were obtained for all 3 materials as seen from the R^2 and Q^2 values (Table 4). Plots of correlation loadings are shown in Fig. 7.

For NP0, 66% of the variation in the predictor variables accounted for 80% variation in MFA (Fig. 7A). Differential pressure had the strongest positive correlation with MFA as seen from their close proximities to each other. MFA was less affected by feeder velocity.

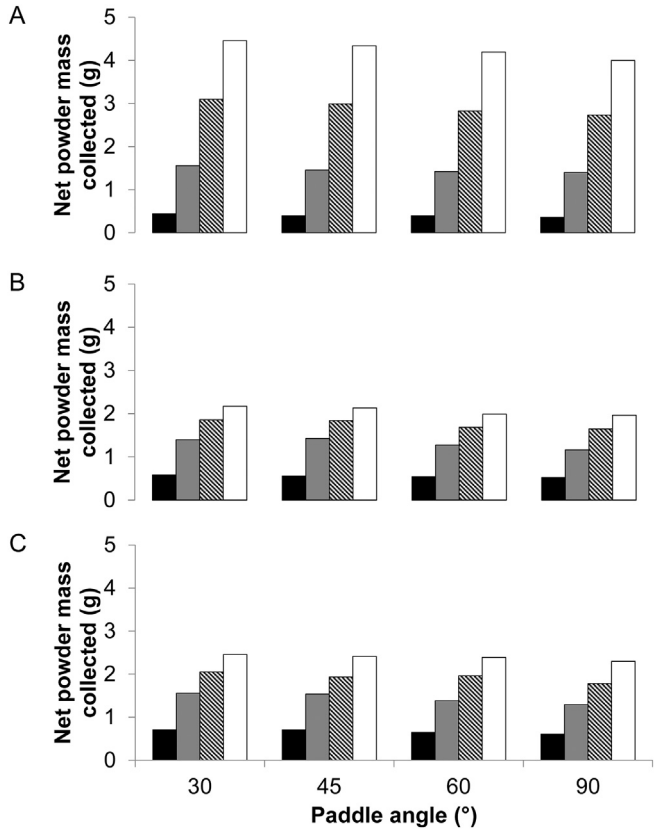


Fig. 5. Pooled average powder masses collected for (A) NP0, (B) T100, and (C) T100M across different paddle angles. Orifice diameters: (■) 2 mm, (▩) 4 mm, (▨) 7 mm and (□) 10 mm.

Table 4
R^2 and Q^2 values from PLSR models.

	MFA_{10}		MFA_7		MFA_4		MFA_2	
Materials	R^2	Q^2	R^2	Q^2	R^2	Q^2	R^2	Q^2
NP0	0.75	0.75	0.78	0.78	0.83	0.83	0.82	0.82
T100	0.75	0.74	0.76	0.75	0.77	0.76	0.81	0.81
T100 M	0.72	0.71	0.71	0.70	0.63	0.62	0.82	0.82

Paddle angle had the least impact on MFA since it was located within the inner ellipse, implying that it accounted for < 50% variation in the dataset. Similarly, for T100 and T100 M, 66% of variation in the predictor variables accounted for 72–77% variation in MFA (Fig. 7B–C). Trends observed were generally similar to those for NP0 and results from the correlation loadings plots thus corroborated those from the main effects plots (Fig. 4).

Weighted regression coefficients for the process variables are presented in Fig. 8. Regression coefficients give an indication of the predictor variables on a response variable. Variables with a large regression coefficient typically have important impact on the response; a positive coefficient indicates a positive link with the response and the reverse is true for a negative coefficient. Predictors with small coefficients are usually negligible in the regression model.

Across all 3 materials, MFA was positively correlated with feeder velocity and differential pressure. In contrast, paddle angle did not significantly affect MFA. Differential pressure had the greatest effect on MFA, suggesting that suction had the highest impact on die fill.

Feeder velocity had greater impact on MFA of T100 and T100M (Fig. 8B–C) as compared to NP0 (Fig. 8A) where its effect increased with increasing orifice diameter. For NP0, the effect of feeder velocity on MFA was not affected by orifice diameter. Differential pressure generally had greater effect on MFA at smaller orifice diameters, especially for T100 and T100M.

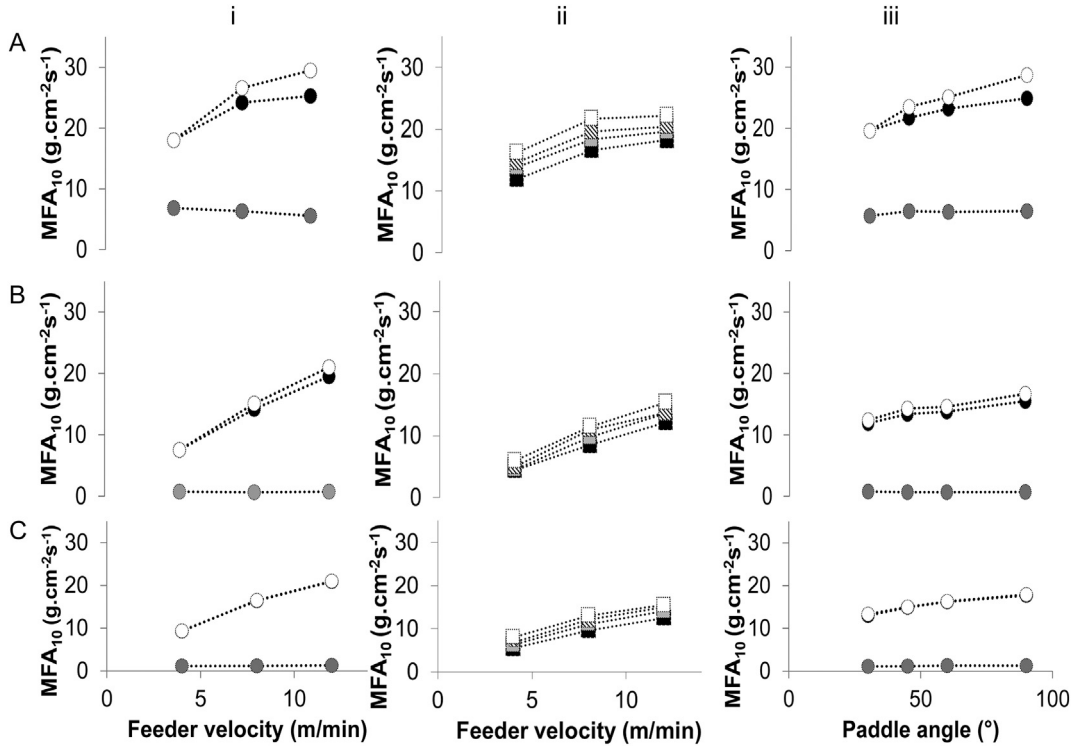


Fig. 6. Two-way interactions between (i) feeder velocity and differential pressure, (ii) feeder velocity and paddle angle and (iii) paddle angle and differential pressure on MFA_{10} of (A) NP0, (B) T100, and (C) T100M. Differential pressure: (●) 0 bar, (●) 0.1 bar and (○) 0.2 bar. Paddle angle: (■) 30°, (▩) 45°, (▨) 60° and (□) 90°.

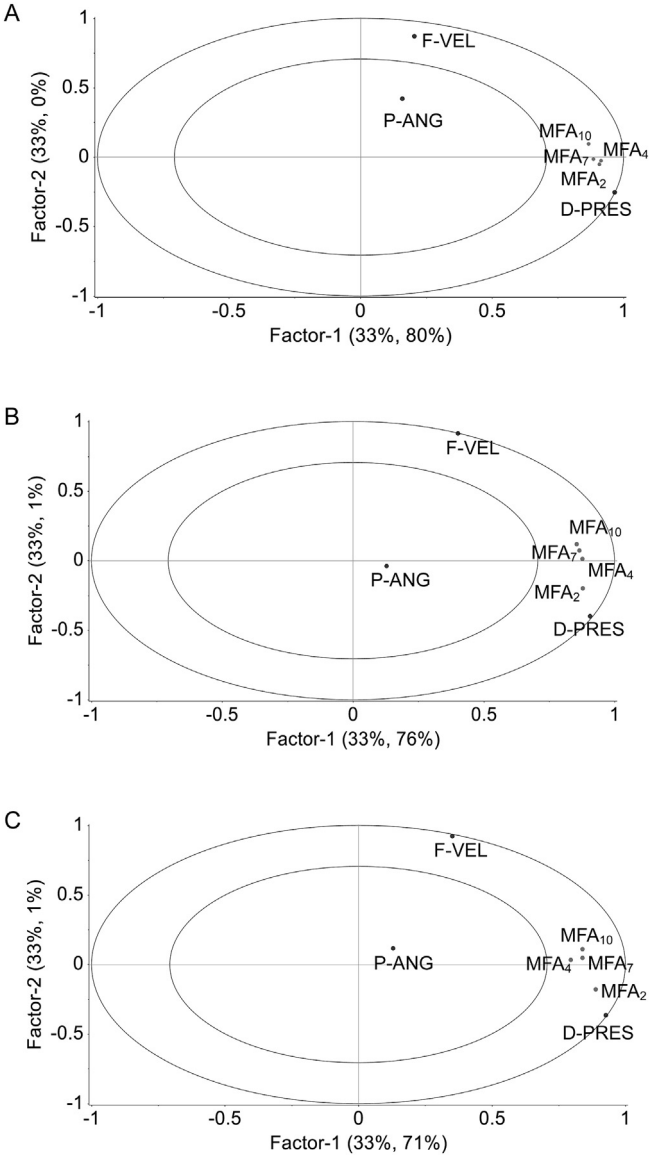


Fig. 7. Correlation loadings plots in the PLSR models for (A) NP0, (B) T100 and (C) T100M. F-VEL: feeder velocity. P-ANG: paddle angle. D-PRES: differential pressure.

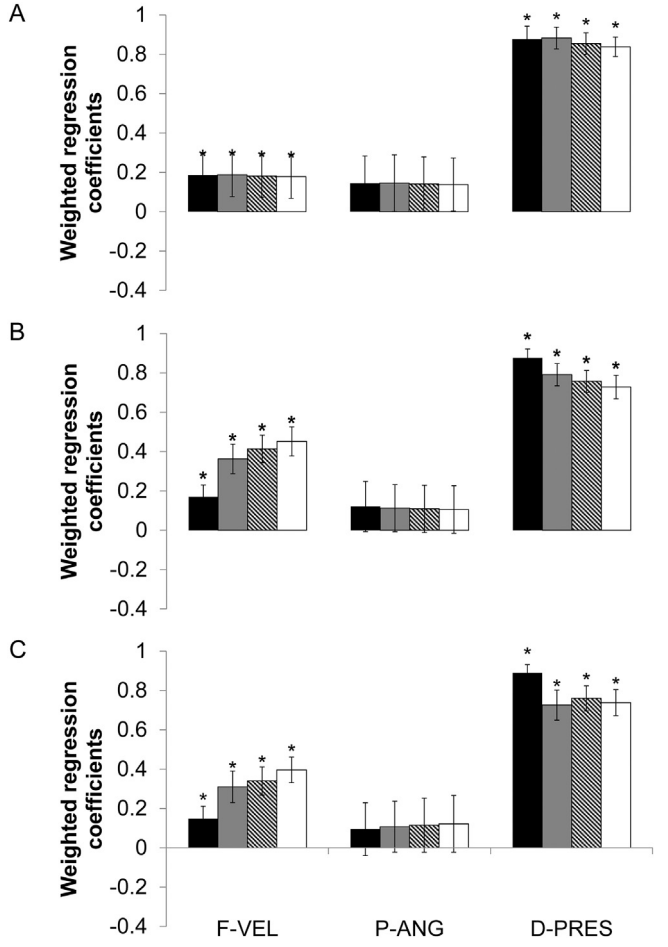


Fig. 8. Weighted regression coefficients of paddle angle, feeder velocity and differential pressure on MFA of (A) NP0, (B) T100 and (C) T100M. MFA: (■) MFA_2, (▒) MFA_4, (▨) MFA_7 and (□) MFA_{10}. Error bars indicate confidence intervals. * indicate significant factors at a 0.05 level.

4. Discussion

4.1. Feeder velocity and die fill

The feeder apparatus used in this study was shown to be able to simulate die filling mechanism at different linear velocities. On an actual rotary press feed frame, the movement of the paddles are not linear but instead angular since the paddle feeders are rotating. This results in a distribution of particle velocities in the feed frame even when the rotational speed of the paddle is constant: particles closer to the hub of the paddle wheel move at lower linear velocities than those further away (Ketterhagen, 2015). The linear movement of the feed unit in this study thus represented part of the distribution of particle velocities possible with a rotating paddle.

The higher the feeder velocity, the higher would be the speed of powder movement across the die and hence, lesser time for die fill to occur. Consequently, the extent of die fill was expected to decrease with increasing feeder velocity. On the contrary, it was found that feed velocity was positively associated with die fill regardless of material or orifice diameter. This could be attributed to the strong interaction between feeder velocity and differential pressure (Fig. 6A(i)–C(i)). As the powder was moved across the platform by the feeder, it was also compressed to some extent towards the back of the trench due to friction with the platform and inertia of the powder bed (Wu et al., 2003). Besides reducing the time for die fill, an increase in feeder velocity would also accentuate the powder compression effect in the trench. Under gravity fill condition (0 bar differential pressure), increase in feeder velocity resulted in slight decrease in MFA of NP0 as a result of reduced time for die fill to occur. MFA of T100 and T100M were less sensitive to increase in feeder velocity since they had much poorer flow properties (Table 2 and Fig. 3) and only very little powder had entered the die orifice even at the lowest feeder velocity. However, under suction fill conditions (0.1 and 0.2 bar differential pressures), the tighter packing of the compressed powder particles facilitated suction fill, hence achieving a better die fill performance despite lesser time for die fill.

Feeder velocity was also found to have considerable effect on die fill of T100 and T100M (Fig. 8B–C) as compared to NP0 (Fig. 8A). Similarly, this could be attributed to the interaction between feeder velocity and differential pressure. T100 and T100 M were more compressible than nonpareils as evidenced from their higher HRs. The higher compressibility accentuated the effect of suction fill in these powders. Furthermore, feeder velocity had greater impact on die fill for larger diameter orifices in T100 and T100M. The larger the orifice diameter, the greater the orifice cross-sectional area exposed to the powder bed per unit time as the powder bed traversed across the orifice.

Consequently, the accentuation of suction fills at increased velocities might have been more effective for larger diameter orifices. The results suggest that the combination of suction fill with high feeder velocity could help to achieve better die fill performance especially for large orifices and compressible powders.

4.2. Forced feeding and die fill

In order to simulate the effect of forced feeding by paddles in feed frames of rotary tablet presses, paddle inserts with different angles of inclination were used in this study. The more acute the angle of inclination, the greater the extent of forced feeding expected since the downward component of force exerted on the powder bed would be greater. Consequently, better die fill performance or higher MFA values were expected with smaller paddle angles. However, paddle angle did not have significant impact on die fill performance regardless of material type as seen from the pooled average powder masses (Fig. 5) and weighted regression coefficients from the PLSR models (Fig. 8).

Smaller paddle angles had poorer die fill efficiency when considering MFA which was normalised against trench volume and time of contact between the powder and the die orifice. Surface area of contact between the paddle and the powder increased with decreasing paddle angle (30° - 538 mm^2, 45° - 399 mm^2, 60° - 337 mm^2 and 90° - 300 mm^2). When the feeder unit moved and pushed powder across the die orifice, the force from the feeder unit was dissipated across a larger number of particles with smaller paddle angles due to the greater area of contact with the powder. This could have reduced the overall forced feeding effect with small paddle angles and thus reduced die fill performance.

4.3. Suction effect and die fill

On rotary tablet presses, suction fill is achieved by the rapid descent of the lower punch at the fill cam to create a partial vacuum which draws feed powder into the die. One method to increase the suction effect on rotary presses is to increase the turret speed as it would bring about a faster descent of the lower punch. However, the fill time available may be compromised as a result. As the designed apparatus in the present study utilises a vacuum pump to simulate the suction fill, it was possible to examine the effects of suction on die fill performance independently, without confounding the time taken for die filling.

Across all the materials studied, differential pressure had the strongest positive effect on die fill which was expected since suction fill is known to increase die fill efficiency (Jackson et al., 2007; Mills and Sinka, 2013; Schneider et al., 2007). Additionally, it was also found that suction effect had greater impact on die fill into smaller orifices as shown by the main effects plots (Fig. 4) and weighted regression coefficients (Fig. 8). As orifice diameter decreases, die fill via gravity becomes even less efficient since there is less available surface area for residual air to escape from the die prior to powder entry. For powders such as T100 and T100M which exhibit arching and ratholing behaviours (Fig. 3), gravity fill through narrow orifices becomes even more challenging or near impossible. Suction effect would be required to initiate and maintain steady flow for these powders especially in narrow orifices, which is in line with the findings by other researchers (Baserinia et al., 2016).

Interestingly, the effect of suction on T100 and T100 M for narrow die orifices was so pronounced that they had higher overall mean MFA_2 and MFA_4 values than NP0, despite possessing poorer flow properties. This could be attributed to the order of powder permeability: NP0 > T100 > T100M. NP0 was the most permeable because of its large particle size and narrow size distribution which created more void spaces in between the particles for the passage of air. On the other hand, addition of fine magnesium stearate particles to T100 M could have filled the void spaces between the lactose particles and reduced its permeability. The less permeable the powder, the greater the effect of suction on die fill. Die fill of coarse and free flowing powders was less affected by suction because air could pass through this powders with relative ease (Schneider et al., 2007; Wu and Cocks, 2004).

While it is known that particle size to die diameter ratio could affect mass flow rate especially for narrow die orifices (Flemming and Mielck, 1995; Kachrimanis et al., 2005), the size fraction of NP0 was much smaller than the narrowest 2 mm orifice. It was thus unlikely that the larger particle size of NP0 could have contributed to particle jamming and adversely affected its die fill relative to T100 and T100 M as seen from the mass flow of NP0 through the narrowest 2 mm orifice (Fig. 3).

Wall effects might also have been more significant for the smaller orifices since a greater part of the powder flow occurs in close proximity to the wall, creating drag forces which could retard powder flow (Kachrimanis et al., 2005). However, given that the depth of the orifice was only 20 mm, it was unlikely that wall effects contributed significantly to the powder flow behaviour.

It follows that under conditions where suction fill is favoured, for example when the orifice is narrow, powder flow properties alone may not be indicative of die fill performance.

5. Conclusion

Die filling, while seemingly simple, is a sophisticated process affected by both powder physical properties and process parameters. To date, the shoe-die model comes closest at simulating actual die fill conditions on a tablet press but lacks the flexibility of different die configurations and forced feeding. A special die filling apparatus was designed in this study in order to overcome these limitations. The apparatus was shown to be capable of generating reproducible measurements as well as simulating passive die filling with added effects of forced feeding and suction fill. The present study evaluates the die fill of pharmaceutical powders with different flow properties using the special apparatus at various process configurations.

Suction effect had the greatest influence on die fill performance. High feeder velocity improved die fill performance, particularly for compressible powders and large orifice diameters, due to interaction with suction effect. Die fill performance in small orifice diameters were more dependent on suction effect, which was in turn affected by powder permeability, due to ineffective gravitational fill at these diameters. Contrary to what was expected, forced feeding had no significant impact on die fill efficiency. Collectively, the results suggest that the relationship between process parameters and die fill performance was highly dependent on the material and orifice diameter. Findings from this study have given a better insight into the process parameters that could affect the die fill process.

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